

Snapdragon Sentinel AI

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Summary

Snapdragon Sentinel AI is an edge-native, privacy-first semantic document retrieval engine engineered specifically for Qualcomm Snapdragon X Elite and Snapdragon series processors.

By leveraging the Qualcomm Hexagon NPU via DirectML, Sentinel AI executes ONNX vector embeddings directly on local hardware. This enables real-time, air-gapped semantic search across local document repositories eliminating reliance on external cloud LLMs, third-party APIs, and internet connectivity.

Key Technical Innovations

- On-Device Vector Embeddings
- Generates high-dimensional vector representations locally using a quantized `text_embedder.onnx` model, ensuring minimal memory footprint and zero cloud latency.
- DirectML Acceleration Stack
- Offloads heavy tensor operations directly onto the Qualcomm Hexagon NPU using ONNX Runtime's `DirectMLExecutionProvider`, bypassing CPU bottlenecks.
- 100% Air-Gapped Data Privacy
- Guarantees zero data leakage, zero telemetry, and zero subscription costs by executing all similarity metrics and text processing 100% offline (0 KB network payload).
- Hardware Latency Benchmarking
- Includes built-in performance telemetry comparing NPU acceleration against CPU execution:
 - NPU Execution: ~780 ms
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 - CPU Fallback: ~2650 ms
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 - Performance Gain: ~3.4x speedup

System Architecture

Layer	Technology	Key Responsibility

Frontend / UI	Streamlit, Plotly	Interactive search dashboard, real-time keyword highlighting, and latency analytics charts.
Inference Engine	SnapdragonNPUEngine	Handles ONNX model binding targeting DirectMLExecutionProvider.
Vector & Retrieval	NumPy, Custom Cosine Math	Computes cosine similarity scores and extracts Top-\$K\$ relevant document chunks.

Impact & Target Use Cases

Designed for high-compliance environments requiring strict adherence to GDPR, HIPAA, and corporate data governance standards:

- Legal Counsel: Instantaneous discovery across confidential contracts and case files without cloud exposure.
- Healthcare & Medical Research: Air-gapped indexing of patient records and restricted biomedical papers.
- Enterprise & Defense: Secure, local document intelligence in low-bandwidth or zero-trust air-gapped networks.